

At Risk Patient Logic: Emergency Triage Level Grouping

In an emergency triage, the goal is mainly to rapidly determine how urgently a patient needs attention. The following logic uses multiple vital signs to assign each patient to a risk level. The following pseudocode allows the medical staff to prioritize who is seen first.

FOR each patient arriving at the ED:

 GET their vital signs:

 GCS, SBP, MAP, pulse, RR, Temp, FiO2

 SET risk_score = 0

 // Consciousness check

 IF GCS < 8:

 risk_score += 3 // severe neurological impairment

 ELSE IF GCS < 13:

 risk_score += 1 // mild to moderate impairment

 // Blood pressure check

 IF SBP < 90:

 risk_score += 2 // hypotensive/possible shock

 ELSE IF SBP > 180:

 risk_score += 1 // hypertensive crisis risk

 // Organ perfusion check

 IF MAP < 60:

 risk_score += 3 // organs not receiving enough blood

 // Heart rate check

 IF pulse < 40 OR pulse > 130:

 risk_score += 2 // dangerous bradycardia or tachycardia

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// Breathing check
IF RR < 8 OR RR > 30:
    risk_score += 2    // respiratory distress

// Temperature check
IF Temp < 35 OR Temp > 39.5:
    risk_score += 1    // hypothermia or high fever

// Oxygen support check
IF FiO2 > 60:
    risk_score += 2    // requires high flow oxygen or ventilator

// Assign the triage level based on risk_score
IF risk_score >= 6:
    LEVEL = "Immediate intervention needed"
ELSE IF risk_score >= 3:
    LEVEL = "See within 15 minutes"
ELSE IF risk_score >= 1:
    LEVEL = "Monitor patient"
ELSE:
    LEVEL = "Not urgent"

RETURN triage level and risk score for each patient

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This is a simple if-else pseudocode which uses abnormal vital signs to determine the required urgency.

For example: A patient with a slightly elevated heart rate is not necessarily critical, but a patient with a low GCS, low MAP and high FiO2 simultaneously is flagged immediately.

To improve this system further, the patients age would be taken because it is an important weighting factor; elderly patients often present subtler vital sign changes despite serious underlying condition. A time dimension would also be clever to introduce to track how vitals

change over the first 30 minutes after arrival, this allow the system to flag patients who are deteriorating even if their initial readings appeared stable.

Pain score and the hospital's risk score threshold will also be important factors to determine the patient's triage level.

References

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